

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Provided technical input and leadership on **complex power system analysis** projects for enterprise clients, translating requirements into actionable deliverables.
- Conducted in-depth **power systems analysis** including **short circuit**, **power flow**, and **protection coordination** studies for a portfolio of energy sector clients.
- Prepared detailed technical and non-technical **reports** for project stakeholders, summarizing findings from **power system studies** and **electrical safety analyses**.
- Managed project expectations and client satisfaction through effective communication and timely responsiveness during remote client meetings.
- Mentored and coached junior engineers on **power system analysis** techniques and best practices, facilitating their technical development.
- Contributed to internal workshops focused on enhancing **company growth** strategies and expanding the international portfolio of services.
- Containerized several analysis tools using **Docker** and deployed them to a managed **AWS EC2** environment, improving accessibility for the engineering team.

SKILLS

Power Systems Engineering – Power System Analysis, Short Circuit Analysis, Power Flow Analysis, Protection Coordination, Arc Flash Analysis, Electrical Safety Analysis, Grid Connection Analysis, Transient Stability, Harmonics, Flicker, Power Quality Evaluation, Earthing, Transformer Inrush, Reliability Assessments, Failure Investigations.

Power System Study Software – SKM, EasyPower, ETAP.

Protection Relays – Bus Differential (87B), Transformer Differential (87T), Restricted Earth Fault, Directional Protection (67).

Standards & Calculations – IEC 60502-2, IEC 60364-5-52, Cable Sizing Calculations.

Cloud & DevOps – AWS EC2, Docker, Jenkins, Git.

Programming & Scripting – Python, SQL, Java, Spring Boot.

Reporting & Analysis – Power BI, Tableau, R, pandas, NumPy.

PROJECTS

Strategic Power System Redundancy Analysis

- Performed strategic **power system analysis** for a major data center client, focusing on redundancy strategies to ensure high availability.
- Conducted detailed **short circuit** and **power flow** studies using **ETAP** to identify potential failure points under various load conditions.
- Generated comprehensive technical reports detailing recommended system upgrades and configuration changes, achieving stakeholder sign-off on proposed solutions.

Arc Flash and Electrical Risk Assessment Automation

- Developed Python scripts to automate parts of the **arc flash** and **electrical risk assessment** process, reducing manual data entry time by an estimated 30%.
- Integrated **Power BI** to visualize key risk indicators and generated standardized reports from the automated analysis outputs for internal review.
- Collaborated with senior engineers to validate script outputs against manual calculations and ensure compliance with internal quality standards.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)